

FRANDEL WANJAWA

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PROJECTS

WasteWise App | *UNEP Hackathon 2025 – Semi-Finalist*

[Link](#)

- Team Lead for the development of an AI- and GIS-powered waste management app.
- Built features to map illegal dumping hotspots and enable recycling exchanges for communities and city authorities.

Agri X – Satellite Bench Model

[Link](#)

- Designed and built a bench model of a nanosatellite to demonstrate satellite-based agricultural monitoring.
- Implemented sensors for soil profiling, climate data collection, and solar energy optimization.
- Focused on remote sensing and IoT integration for precision agriculture.

Tafiti CubeSat – Engineering Model

[Link](#)

- Contributed to the development of an engineering model of the Tafiti CubeSat.
- Worked on the communication subsystem and embedded systems integration for space-grade reliability.
- Assisted in hardware-software interfacing and testing to validate CubeSat design.

MedicApp

[Link](#)

- Developed a mobile health application tailored for transparent hospitals.
- Designed to ensure timely processing of insurance claims and improve hospital–insurer workflows.
- Enhanced accountability, efficiency, and transparency in healthcare financial management.

Mailsasa

- Developed an automated mail sorting and dispatch system for the Telecommunications Compliance Department at the Communications Authority of Kenya, streamlining the delivery of 1,000+ emails and reducing processing time from one week to a few hours.

EDUCATION

Jomo Kenyatta University of Agriculture and Technology (JKUAT), Juja, Kiambu, Kenya

Bachelor of Science in Electronic and Computer Engineering, June 2025.

Second Class Honors (Upper Division).

Friends School Kamusinga, Kimilili, Bungoma, Kenya

Kenya Certificate of Secondary Education, 2018.

A-(minus).

The King David Preparatory-Kolanya, Kolanya, Teso-North, Busia, Kenya

Kenya Certificate of Primary Education, 2014.

400 marks.

EXPERIENCE

Communication Authority of Kenya

Compliance and Licensing Engineer

Jan, 2025 – Present

- Design and develop internal software tools to automate compliance and licensing workflows within the Telecommunications Regulatory framework.
- Digitize and optimize regulatory processes, reducing manual handling, processing delays, and operational bottlenecks.
- Build workflow automation systems and tracking dashboards to improve visibility of application status, compliance reviews, and approvals.
- Collaborate with compliance officers and licensing teams to translate regulatory procedures into scalable digital solutions.
- Implement data validation and process control mechanisms to improve accuracy and consistency in license and compliance assessments.
- Support integration of digital systems with existing regulatory databases and platforms.

ICT Authority- Kenya

AI & Software Engineer

Dec, 2026 - Present

- Design and develop AI-driven and software solutions to improve efficiency, transparency, and accessibility of public service delivery.
- Lead digital innovation initiatives by identifying manual government workflows and transforming them into intelligent automated systems.

- Build and deploy data-driven tools that support evidence-based decision-making across public sector departments.
- Collaborate with cross-functional government teams to translate policy and operational needs into scalable technical solutions.
- Optimize service delivery processes through automation, analytics, and machine learning models that reduce turnaround times and operational costs.

UNEP Hackathon 2025 – Semi-Finalist

Team Lead

Aug 2025

- Led the WasteWise team in designing and developing an AI- and GIS-powered app to improve waste management for local communities and city authorities.
- Directed solution design to map illegal dumping hotspots and create a platform for individuals to offer recyclable items, fostering responsible disposal practices.
- Applied AI-driven analytics and geospatial intelligence to support better decision-making and reduce environmental pollution.
- Completed hackathon masterclasses on AI, GIS, and sustainability, integrating key learnings into the solution.

Tethics Electrics Group

Co-Founder

Apr, 2025 - Present

- Building diverse innovative solutions in the fields of AI, ML, and IoT.
- Sustaining a strong focus on climate action and sustainability.
- Coordinated the design and development of AI-powered embedded systems for environmental monitoring.
- Pioneered the integration of AI and IoT technologies in climate-smart agriculture systems to optimize irrigation, soil monitoring, and crop yield forecasting.

Kenya Space Agency

TT&C Subsystem Lead

Jan, 2024 – Mar, 2025

- Established a suitable implementation framework and work plan for the TT&C subsystem.
- Designed and Implemented a data visualization tool for the satellite data.
- Lead, mentor, and manage a high-performing engineering students' team, fostering a collaborative and results-driven work environment.
- Co-designed and implemented the communication circuitry for a nano satellite and proofed firmware for the communication subsystem.
- Resolved a preexisting backlog in the TT&C subsystem and ended up delivering a working model.

Netgenix Tech Icons

CTO

Aug, 2023 – July, 2024

- Developed and implemented a comprehensive technology strategy to drive innovation and growth within the company.
- Laid the foundation for a robust technology infrastructure, ensuring scalability and reliability.
- Led the design and development of cutting-edge software solutions, leveraging emerging technologies to enhance product quality and user experience.

EgyPro EA Limited

Engineering Intern

Jan, 2023 – May, 2023

- Achieved real-time mitigation of electrical and mechanical faults in booster stations owned by Safaricom in Kiambu County.
- Ensured seamless transmission of telemetry data to the Egypro command center.
- Determined the most effective solutions to address the needs of various booster sites.

Gearbox EA || Fedha Electrics || Untapped Global

Engineering Intern

Jan, 2022 – May, 2022

- Co-developed smart solutions addressing human challenges, including a smart water hub and speed limiter.
- Developed data dashboards for IoT sensors.
- Co-developed firmware for hardware designed by other engineers.

Kenya National Bureau of Statistics

Census Enumerator

July, 2019 – Aug, 2019

- Collected and recorded data on the population of Kenya during the 2019 National Population and Housing Census.
- Ensured accuracy and completeness of data collected.
- Provided support to other Census enumerators.

PRESENTATIONS

1. TT&C Subsystem best practices:

Organizer: Women in Engineering - Multimedia University, Kenya.

2. TT&C subsystem in and out:

Organizer: Cubekraft, Kenya.

CERTIFICATIONS

1. **Python Essentials I & II** -Cisco
2. **JavaScript Essentials I&II**-Cisco
3. **English for IT**-Cisco
4. **Creating Compelling Reports**-Cisco
5. **Digital Awareness**-Cisco
6. **Getting Started with Artificial Intelligence** – IBM
7. **Introduction to Cybersecurity** – Cisco
8. **Robotics Dojo** – JKUAT-JICA
9. **Build your first chatbot** – IBM
10. **Microsoft AI Fluency Course** – Microsoft
11. **Cloud for CTOs** – AWS
12. **Getting Started with DevOps on AWS** – AWS
13. **Project Management Foundations** – LinkedIn Learning
14. **Career Essentials in GitHub Professional Certificate** – GitHub
15. **Cybersecurity Awareness: Ransomware** – LinkedIn Learning
16. **Delivering an Authentic Elevator Pitch** – LinkedIn Learning
17. **Learn Emotional Intelligence: The Key Determiner of Success** – LinkedIn Learning
18. **Learning to be Approachable** – LinkedIn Learning
19. **Unconscious Bias** – LinkedIn Learning
20. **Communicating Values** – LinkedIn Learning
21. **ETL in Python and SQL** – LinkedIn Learning
22. **Holding Yourself Accountable** – LinkedIn Learning
23. **Persuading Others** – LinkedIn Learning
24. **Complex Negotiation Tips** – LinkedIn Learning
25. **Engineering as a Path to Leadership** – Society of Engineering Students
26. **Cubesat Design and Development** – Italian Space Agency
27. **Altium Education PCB Design Course** – Italian Space Agency
28. **Cities and Climate Change** – UNITAR
29. **Introduction to Machine Learning** – Kaggle
30. **Python Coder** – Kaggle
31. **Python (Basic)** – Hackerrank
32. **Climate Adaptation and the Cost of Inaction** – LinkedIn Learning
33. **Climate Justice and Sustainable Business Practices** – LinkedIn Learning
34. **Diversity, Inclusion and Belonging** – LinkedIn Learning
35. **Overcoming Procrastination** – LinkedIn Learning
36. **Problem-Solving Techniques** – LinkedIn Learning
37. **Smart Cities: Solving Urban Challenges Using Technology** – LinkedIn Learning
38. **Navigating Environmental Sustainability: A Guide for Leaders** – LinkedIn Learning
39. **Run leader (Organizing campaigns)** – Urban better
40. **Run Tracker (Contributing to data collection)**- Urban better

SKILLS

- Software design and development
- Process Automation
- Complex negotiations
- Printed Circuit Board (PCB) Design
- Purchasing Processes
- Climate Change Mitigation and Adaptation
- Artificial Intelligence (AI)

- Machine Learning
- Natural Language Processing (NLP)
- IBM Watson
- Circuit Design
- Web Development
- Firmware Development
- Project Management

REFEREES

Name: John Omwamba

Referee's position: Urban Planning Officer

Email: <mailto:john.omwamba@un.org>

City: Trieste

Country/territory: Italy

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Name: Njeru Manegene

Referee's position: Lecturer ECE JKUAT

Email: <mailto:njeru.manegene@jkuat.ac.ke>

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PEER MENTOR

Name: Ian Toyota

Mentor's position: Student Harvard University

Email: <mailto:iantoyota@college.harvard.edu>

City: Cambridge, Massachusetts.

Country/territory: United States of America.

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